CLASS 10 — SCIENCE · EASY LEVEL UNIT TEST SET 2

Chapters 9–12: **Light - Reflection and Refraction, The Human Eye and the Colourful World, Electricity, Magnetic Effects of Electric Current** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Easy level: direct, single-step questions.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. The SI unit for the strength of the magnetic field is the 1

(a) henry **(b)** weber only **(c)** ampere **(d)** tesla

2. A ray of light going from air to water bends 1

(a) randomly **(b)** towards the normal

(c) not at all **(d)** away from the normal

3. A person who cannot see nearby objects clearly is 1

(a) astigmatic **(b)** hypermetropic **(c)** myopic **(d)** blind

4. Thicker wires have 1

(a) less resistance **(b)** the same resistance

(c) more resistance **(d)** infinite resistance

5. Oersted observed that a compass needle moves near a wire carrying 1

(a) an electric current **(b)** sound

(c) light **(d)** no current

6. A convex lens is thicker at the 1

(a) edges **(b)** centre **(c)** both equally **(d)** none

7. The band of seven colours formed after dispersion is the 1

(a) shadow **(b)** rainbow only **(c)** spectrum **(d)** beam

8. Two resistors of 4 ohm each in parallel have a total of 1

(a) 2 ohm **(b)** 16 ohm **(c)** 8 ohm **(d)** 4 ohm

9. In Fleming's left-hand rule, the forefinger indicates 1

(a) the voltage **(b)** the force **(c)** the current **(d)** the field

10. In reflection, the angle of incidence is equal to 1

(a) the angle of reflection **(b)** the angle of refraction

(c) zero **(d)** 90 degrees

11. The coloured part of the eye is the 1

(a) pupil **(b)** iris **(c)** retina **(d)** cornea

12. The instrument used to measure current is the 1

(a) voltmeter **(b)** galvanometer only
(c) thermometer **(d)** ammeter

13. Like poles of magnets 1

(a) merge **(b)** do not interact
(c) attract each other **(d)** repel each other

14. The speed of light is maximum in 1

(a) water **(b)** vacuum **(c)** diamond **(d)** glass

15. Spectacles for myopia use 1

(a) concave lenses **(b)** prisms **(c)** plane lenses **(d)** convex lenses

SECTION — B (Very Short Answer Type) 5 × 2 = 10

Answer in about 20–30 words each.

16. Name the defect of vision corrected by a concave lens. 2

17. Define the resistance of a conductor. 2

18. State the right-hand thumb rule. 2

19. Which has more refractive index, glass or water? 2

20. Name the colour that is deviated the most and the least by a prism. 2

SECTION — C (Short Answer Type) 5 × 3 = 15

Answer in about 40–60 words each.

21. State Fleming's left-hand rule. 3

22. State the lens formula and the formula of magnification. 3

23. Why is the sky dark for an astronaut? 3

24. Convert 1 kWh into joules. 3

OR

Name the three wires in a domestic circuit with their colours. 3

25. Write the laws of reflection of light. 3

OR

State three differences between myopia and hypermetropia. 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

26. An object is placed 40 cm from a concave mirror of focal length 20 cm. Find the image position and nature. 5

OR

Explain how a rainbow is formed. 5

27. A bulb rated 100 W at 220 V is used for 10 hours. Find the energy consumed in kWh and the current drawn. 5

OR

Describe the principle, construction and working of an AC generator. 5

INTERMIND

END OF QUESTION PAPER **Total: 50 Marks**

MASTERMIND